EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A virtual machine has its own wholly separate physical processor.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





DO NOW — RECALL

# BIOS, drivers and virtual machines

Name: _____ Class: _____ Date: _____

**1. What does an OS driver do?**

**2. Define multi-user OS.**

**3. What executes before the OS is loaded?**

**4. Simplify (¬A∧B)∨(A∧B). Show factoring.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

Interfaces with specific hardware.

**2. Model answer**

Supports separate user sessions and permissions.

**3. Model answer**

Platform firmware and boot-loading code.

**4. Model answer**

B∧(¬A∨A)=B∧1=B.

**Vocabulary — know these terms**

UEFI — Modern system boot firmware.

driver — Hardware-specific software interface.

VM — Virtual machine.

hypervisor — VM hosting software.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 3**

Evaluate guest virtual machines for school testing of unfamiliar operating systems. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

State roles of UEFI/BIOS, boot loader, driver and hypervisor. [4]

### Question 2

Explain how a compatible printer driver enables printing. [6]





NEW LEARNING

# Read, annotate and explain

### Understand

BIOS/UEFI firmware initialises key hardware and starts the boot process.

### Trace

Device drivers connect OS abstractions to the instructions and data expected by actual peripherals.

### Evaluate

A hypervisor provides virtual machines for isolation, testing and consolidation, though CPU/RAM/storage overhead remains.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**A printer is connected but software cannot print; name an OS dependency.**

### Predict first

A missing or incompatible driver can prevent correct communication with a device.

### Model and check

Explain why each value follows from the earlier step.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve